

6-Month AI Upskilling Roadmap for Data Engineers

From Data Engineer → AI/ML Engineer · MLOps · AI Data Architect

2+ Years XP · 6 Months · Career Switch Ready

6	3	12+	40%+	2
Months	Target Roles	Projects	Salary Jump	Certifications

Why This Plan?

You already have solid data engineering fundamentals — pipelines, SQL, cloud infra. The AI era rewards engineers who can **build, deploy, and operate ML/AI systems**, not just move data. This plan layers AI/ML engineering skills on your existing stack in 6 focused months, positioning you for roles that combine higher pay with better WLB.

Role	Why It Fits You	Avg. Salary (India)	WLB
ML Engineer	Pipelines + model serving	■18–35 LPA	■■■■■
MLOps / AI Infra Eng.	Your DE skills = direct transfer	■20–40 LPA	■■■■■
AI Data Architect	Senior, design-led, remote-friendly	■25–50 LPA	■■■■■

MONTH 1

ML Foundations + Python Upgrade

■ Build the Base

Goal

Bridge from data engineering to ML. Master the math intuition, Python ML stack, and ship your first end-to-end ML model.

Week 1–2

- Linear Algebra & Statistics refresher (Khan Academy / 3B1B)
- Python: NumPy, Pandas deep-dive, type hints, dataclasses
- Scikit-learn: regression, classification, pipelines
- Feature engineering on a real Kaggle dataset
- SQL optimisation + window functions revision

Week 3–4

- Fast.ai — Practical Deep Learning (Lesson 1–4)
- Jupyter → VS Code + MLflow tracking setup
- Version your data with DVC (links to your DE background)
- Read: Hands-On ML with Scikit-Learn (Ch 1–6)
- Daily: 1 LeetCode (Easy/Medium) — Python

■ Month 1 Deliverable

End-to-end ML project on GitHub: data ingestion → feature store → trained model → evaluation report. Use an existing DE dataset you already know.

scikit-learn

MLflow

DV
C

Pandas

NumP
y

Fast.ai

MONTH 2

Deep Learning + LLM Fundamentals

■ Understand the Engine

Goal

Understand how modern AI works — transformers, LLMs, embeddings — so you can confidently build on top of them and speak the language in interviews.

Week 1–2

- Neural Networks: backprop, activations, loss functions
- PyTorch basics — tensors, autograd, custom datasets
- CNNs + transfer learning (fast iteration via HuggingFace)
- Transformer architecture deep-dive (Andrej Karpathy's videos)
- Embeddings, vector search, cosine similarity

Week 3–4

- HuggingFace: fine-tune a BERT/DistilBERT classifier
- Build a RAG (Retrieval-Augmented Generation) mini-app
- LangChain / LlamaIndex fundamentals
- OpenAI & Anthropic API — prompt engineering patterns
- Read: The Illustrated Transformer (Jay Alammar)

■ Month 2 Deliverable

RAG chatbot over a domain-specific document corpus (e.g., internal wiki or public API docs). Deployed as a simple FastAPI endpoint. GitHub + demo GIF.

PyTorch

HuggingFace

LangChain

LlamaIndex

FAISS

FastAPI

Goal

This is where your Data Engineering background becomes a massive advantage. Build production-grade ML systems — CI/CD for models, monitoring, feature stores.

Week 1–2

- Docker for ML: containerise models, multi-stage builds
- Kubernetes basics (k8s on minikube → EKS/GKE)
- CI/CD for ML: GitHub Actions + model testing
- Feature Store: Feast or Tecton concepts
- Airflow → Kubeflow / Prefect for ML pipelines

Week 3–4

- Model serving: BentoML, Ray Serve, or Triton
- Model monitoring: Evidently AI, Arize, or Whylogs
- Cloud ML: SageMaker OR Vertex AI (pick one deep-dive)
- Terraform basics for ML infra as code
- Study: Google MLOps Whitepaper + Chip Huyen's DMLS Ch 7–9

■ Month 3 Deliverable

Automated ML pipeline: data ingestion → training → evaluation → deployment with drift monitoring. Show CI passing on PR. This is your resume centrepiece for MLOps roles.

Docker

Kubernetes

SageMaker / Vertex AI

Feast

Prefect

Evidently

Goal

Go beyond demos. Build production-quality AI applications — agents, tools, and data pipelines specifically designed for LLM workloads.

Week 1–2

- LLM fine-tuning: LoRA / QLoRA with HuggingFace PEFT
- Agentic AI: LangGraph / CrewAI — multi-agent orchestration
- Vector databases: Pinecone, Weaviate, or pgvector in depth
- Prompt caching, batching, cost optimisation patterns
- AI-native data pipelines (unstructured → structured)

Week 3–4

- Build a full-stack AI app (Next.js + FastAPI + LLM backend)
- Evaluation frameworks: Ragas, DeepEval for RAG quality
- Guardrails AI / NeMo Guardrails for safe deployments
- Study: OpenAI Cookbook + Anthropic prompt engineering guide
- Contribute to one open-source AI/MLOps project on GitHub

■ Month 4 Deliverable

AI agent that solves a data engineering task autonomously (e.g., auto-generates dbt models from schema descriptions, or answers data quality questions). Deploy on cloud with a public URL.

LangGraph

LoRA/QLoRA

Pinecone

Ragas

Next.js

CrewAI

MONTH 5

System Design + Certifications

■ Think at Scale

Goal

Senior engineers are hired for system thinking, not just coding. Learn to design large-scale AI systems AND get certified to signal credibility.

Week 1–2

- ML System Design: design Twitter/Netflix/Uber ML features
- Scalability patterns: batch vs streaming inference, caching
- Data architecture for AI: lakehouse + vector + feature store
- Trade-off discussions: cost, latency, accuracy, explainability
- Study: Designing ML Systems (Chip Huyen) — full book

Week 3–4

- Certification Target 1: AWS ML Specialty OR GCP ML Engineer
- Certification Target 2: Databricks Certified ML Associate
- Mock ML system design interviews (Exponent / Pramp)
- Contribute 2 blog posts to Medium/Substack on your projects
- Review and refine GitHub portfolio — READMEs matter

■ Month 5 Deliverable

At least 1 certification passed. System design doc for a hypothetical AI product (write it like a real RFC). Published blog post with 100+ views.

AWS ML Cert

GCP ML Cert

Databricks Cert

System Design

Technical Writing

MONTH 6

Job Hunt Sprint + Interview Prep

■ Land the Role

Goal

Transition from learning mode to landing mode. Polish your brand, apply strategically, and nail every interview stage.

Applications & Branding

- Resume: tailor for ML Eng / MLOps / AI Data Arch separately
- LinkedIn: update headline + skills + featured projects
- GitHub: all 4 projects pinned, READMEs with architecture diagrams
- Cold outreach: 5 personalised DMs/week to hiring managers
- Target: FAANG, unicorn startups, AI-native product companies

Interview Preparation

- DSA: 2 LeetCode problems/day (medium focus)
- ML interviews: model trade-offs, bias-variance, CV/NLP Qs
- Behavioural: STAR stories around your DE → AI transition
- Take-home challenges: treat each one as a mini-project
- Negotiate: research bands on Levels.fyi + LinkedIn Salary

■ Month 6 Deliverable

20+ applications sent, 5+ first-round interviews completed, offer in hand or pipeline. You have 6 months of compounding proof — show it.

LeetCode

System Design

ML Interviews

LinkedIn

Levels.fyi

Negotiation

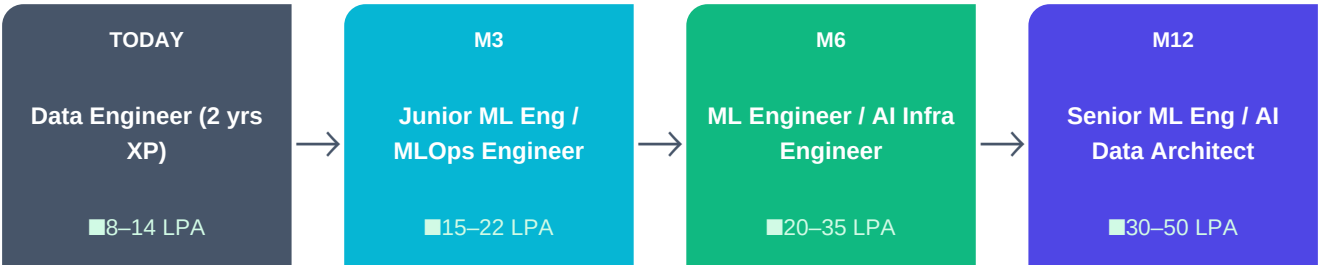
Suggested Weekly Schedule

Day	Morning (1–1.5 hr)	Evening (1–1.5 hr)	Weekend
Mon–Fri	Core concept study (videos / reading)	Hands-on coding / project work	
Saturday	Project build sprint (3–4 hrs focused)	Review + notes	Deep project work
Sunday	Revision + flashcards	LeetCode / mock interview	Rest & planning

Top Resources

Resource	Type	Best For	Cost
Fast.ai Practical DL	Course	DL intuition fast	Free
Designing ML Systems — Chip Huyen	Book	System design	~\$2k
Full Stack Deep Learning (FSDL)	Course	MLOps end-to-end	Free
HuggingFace Learn	Course	Transformers + NLP	Free
Andrej Karpathy YouTube	Videos	Neural net intuition	Free
AWS / GCP ML Cert Prep	Course	Certification	~\$3–5k
Databricks Academy	Course	ML + Data Lakehouse	Free tier
LeetCode (NeetCode roadmap)	Practice	DSA interviews	Free/Pro
Levels.fyi + LinkedIn Salary	Research	Salary benchmarks	Free

Career Progression Roadmap



Pro Tips for Success

■ Leverage Your DE Background

Always frame ML work through data pipelines — you already understand data quality, latency, and scale. This differentiates you from pure ML newcomers.

■ Projects > Certificates

Recruiters for AI roles scan GitHub before LinkedIn. Make every project README tell a story: problem → your solution → impact.

■ Build in Public

Post weekly progress on LinkedIn. Tag tools you use. Even short posts about lessons learnt compound into a visible AI brand over 6 months.

■ Community is a Cheat Code

Join MLOps Community Slack, Hugging Face Discord, and local AI meetups. Referrals account for 40-60% of hires in AI roles.

■ Protect Your Energy

1–3 hrs/day of deep work beats 6 hrs of distracted study. Sleep, exercise, and one full rest day per week are non-negotiable.